nature communications

Article | Article in Press

https://doi.org/10.1038/s41467-026-79083-7

Antibiotic contamination stimulates microbial methylmercury synthesis in rice paddies

Received: 1 April 2026

Accepted: 12 October 2026

Published online: 27 October 2026

Check for updates

Hailong Li[1], Xiangzhen Li[2], Xiao Zhang[1], Wenli Tang[1] ✉, Jingli Liu[3], Ri-Qing Yu[4] ✉, Christopher Rensing[5], Yi Luo[1], Xiaolong Wang[1], Juntao Xia[6], Ya-Nan Yang[1], Yang Hu[1], Hao-Ran Zhao[1], Yujiao Liu[1], Mengjie Wu[1], Chengjun Li[7], Lin Xiao[1] ✉ & Huan Zhong[1] ✉

Antibiotic residues are now widespread in agricultural soils, yet their influence on microbe-mediated mercury (Hg) transformations and associated risks is poorly understood. Here, we demonstrate that fluoroquinolone (FQ) antibiotics enhance the production of neurotoxic methylmercury (MeHg) in paddy soils by selectively altering microbial metabolic activity. In 40 paddy soils from China's major rice-growing regions, FQ addition at 10 mg kg^{-1} increased MeHg content by 2.9%–19.7% after 21 days of incubation. Metabolic inhibition and microcosm experiments revealed that this increase stemmed primarily from stimulated microbial Hg methylation, rather than inhibited MeHg demethylation. Sulfate-reducing bacteria (SRB) dominated this process since inhibiting sulfate reduction decreased FQ-induced Hg methylation by 31%–75%. Integrated microbiome and functional prediction based on soil microbial communities and metagenomic analyses of enriched SRB consortia further identified *hgcA*-carrying SRB as the key methylators. FQ exposure selectively suppressed competing microbial groups while promoting metabolically active, antibiotic-tolerant SRB, particularly those using the reductive acetyl-CoA pathway, thereby elevating Hg methylation activity without significantly changing methylator abundance or community structure. Together, these findings reveal that widespread antibiotic contamination amplifies Hg risks in soils via tolerant microbial communities, potentially representing a hidden yet critical link between emerging contaminants, heavy metals, and food safety.

Fluoroquinolones (FQs) are a class of broad-spectrum antibacterial agents[1] that are effective against a wide range of Gram-negative ($Gram^-$) and Gram-positive ($Gram^+$) pathogenic bacteria[1,2]. They inhibit bacterial replication through the blockage of DNA supercoiling by binding the gyrase-DNA complex before DNA cleavage[3]. Over the past decades, FQs have been extensively used in both human prescriptions and veterinary medicine[4], with approximately 70%–90% of the orally administered FQs excreted[5] and subsequently entering the environment, including farmlands. Particularly, antibiotics accumulate in agricultural soils through pathways such as wastewater irrigation[6] and the application of bio-fertilizers[7,8]. For instance, FQs were detected in all 241 soil sites in the Yangtze River delta[9], a crucial agricultural and industrial region in China. The residue concentrations of two typical FQs, ciprofloxacin (CIP) and enrofloxacin (ENR), reached up to 7.2 and

A full list of affiliations appears at the end of the paper. ✉e-mail: tangwenli@nju.edu.cn; ryu@uttyler.edu; xiaolin@nju.edu.cn; zhonghuan@nju.edu.cn

6.1 mg kg^{-1}, respectively[10,11], in manure-applied vegetable fields. Collectively, these findings demonstrate that agricultural soils now serve as significant sinks for antibiotics.

While the impact of FQ residues on soil microbial communities has been recognized[12], their potential effects on microbial Hg methylators and the production of neurotoxic methylmercury (MeHg) remain unknown. The synthesis of MeHg is mediated by anaerobic microorganisms carrying *hgcAB* genes[13]. Among these microorganisms, sulfate-reducing bacteria (SRB) are widely regarded as a principal Hg-methylating group[14], with over 30 SRB strains having been experimentally confirmed as MeHg producers[15]. Most complete-oxidizing SRB metabolize organics via the acetyl-CoA pathway[16], although MeHg synthesis is not exclusively dependent on it[17,18]. Notably, certain SRB taxa known or putative to methylate Hg, such as *Desulfobacter* and *Syntrophobacter*, were found enriched following CIP treatment in sludge systems[19–21]. A prior study also reported that CIP could stimulate SRB growth in marsh sediments[22], and pure-culture experiments indicated increased MeHg production by *Geobacter metallireducens* GS-15 upon exposure to FQs[23]. We therefore hypothesize that FQs may reshape microbial Hg methylation in paddy soils by altering key microbial processes. Given that MeHg produced in soil can accumulate in rice grains[24], understanding this antibiotic-microbe-Hg nexus is critical. Should FQ-induced shifts in microbial methylation occur in nature, widespread antibiotic residues could ultimately influence dietary MeHg exposure, an established risk factor for neurodevelopmental impairments[25].

Here, we examined the impact of FQ contamination on the microbial production of MeHg in paddy soils. Rice paddies were chosen for their high Hg methylation potential compared to other ecosystems[26], as well as the widespread occurrence of FQ antibiotics in these soils[4,9]. Using 41 soils collected from nine major rice-growing provinces across China, we conducted microcosm experiments to assess how FQ input influences MeHg production. We further investigated the underlying mechanisms through microbial metabolic assays, 16S rRNA gene profiling, and metagenomic sequencing. This study advances our understanding of the interplay between emerging antibiotic contamination and Hg biogeochemical cycling, with direct implications for global food security.

Results

FQ input elevates MeHg contents in paddy soils

To examine the response of key Hg-methylating microbes to antibiotic stress, we selected three soils dominated by various Hg-methylating microbial groups: one with a relatively high abundance of Hg-methylating SRB (*hgcA*-SRB) (designated as MY soil, 31.5%; Supplementary Table 1), one dominated by Hg-methylating FeRB (*hgcA*-FeRB) (74.4%, DL soil), and one dominated by Hg-methylating methanogens (*hgcA*-Methanogens) (39.3%, KL soil)[27]. Each soil was spiked with the four widely used FQs separately, namely ciprofloxacin (CIP), enrofloxacin (ENR), norfloxacin (NOR), and ofloxacin (OFL), at concentrations of 10 μg kg^{-1} and 10 mg kg^{-1} (Batch assay I). At a low dose of FQ addition, minor yet significant increases in soil MeHg were only observed in MY soil ($P < 0.05$; Supplementary Fig. 1). In contrast, a higher spike level of 10 mg kg^{-1} produced clearer soil-specific results. In MY soil, CIP and ENR treatments significantly increased MeHg production compared to the non-spike soil (CK) on day 7, and all four FQs led to significant elevations in soil MeHg on day 30 (Fig. 1a and Supplementary Fig. 2). In DL soil, only the CIP treatment significantly elevated net MeHg accumulation, whereas no significant changes were observed in the KL soil. Additionally, in the MY soil, in which the FQ-induced increases in MeHg were the most notable, only minor changes were observed in soil pH (Supplementary Fig. 3) and bioavailable Hg pools, indicated as dissolved total Hg (Supplementary Fig. 4). These results suggest that FQ treatment exerted minor impacts on abiotic factors in the soil.

To further assess the widespread effects of FQs on microbial MeHg synthesis, we tested 40 paddy soils collected from major rice-planting areas in China (soil properties and location listed in Supplementary Table 2) with FQ treatments (referred to as Batch assay II). After a 21-d incubation, more than 40% of the added ENR remained in soils, indicating that the spiked FQ was not completely degraded (Supplementary Table 3). The overall stimulatory effects of four FQs on soil MeHg synthesis varied across provinces, with provincial average increases in MeHg contents ranging from 2.9 to 19.7%, and the highest provincial increases reached 16.8%, 22.8%, 19.1%, and 25.3% for CIP, ENR, NOR, and OFL, respectively (Fig. 1b–d and Supplementary Tables 4–5). Interestingly, changes in Hg-methylation potentials (defined as the concentration ratio of MeHg to THg, %) under FQ treatments were significantly higher in Southern China than in Northern China ($P < 0.01$; Fig. 1c, d). Such a spatial distribution pattern was not induced by the variation in soil sulfate contents, as the two exhibited different patterns (Supplementary Fig. 5). This suggests that the geographic pattern of FQ influence could be controlled by complex soil biogeochemical properties and microbial community composition.

SRBs are the primary drivers of the elevated MeHg production

Given that FQ significantly elevated Hg methylation potential in the *hgcA*-SRB-dominated MY soil ($P < 0.05$; Supplementary Fig. 2), we hypothesized that *hgcA*-SRB may respond to FQ amendments. To test this hypothesis, four soils (i.e., Huanggang, Tianmen, Meishan, and Ganzhou) exhibiting the highest Hg-methylation potentials and net MeHg increases (139%–196%; Supplementary Table 4) under FQ amendments among 40 tested soils were used in Batch assay III, in which molybdate was applied to inhibit microbial sulfate reduction, a key metabolic process of SRB.

Results showed that without molybdate (Mo-free), FQ consistently enhanced Hg-methylation potentials compared to controls (CK; red bars in Fig. 2a–d). In contrast, molybdate addition, irrespective of the doses applied (5, 10, and 20 mM; Supplementary Text 1), was able to reduce FQ-induced Hg-methylation by 18.6%–77.5% (+Mo, blue bars; Fig. 2a–d) compared to the Mo-free group (red bars; Fig. 2a–d; Supplementary Fig. 6). Stable Hg isotope analyses further showed that addition of molybdate inhibited Hg methylation rate by up to 85.6%, while the impact on MeHg demethylation was minor (Supplementary Fig. 7). These results indicated that SRB activity was essential for the FQ-induced rises in MeHg production. Correspondingly, molybdate markedly inhibited sulfate reduction (Fig. 2e–h), confirming effective suppression of SRB. Meanwhile, molybdate addition also reduced the abundance of *Deltaproteobacterial hgcA* genes ($P = 0.007$; Fig. 2i), while causing only minor changes in Archaeal *hgcA* copies (Fig. 2j), implying that sulfate-reducing *Deltaproteobacteria* were the potential methylators responsible for FQ-amplified MeHg synthesis. Furthermore, significant correlations were found between MeHg increase rates and sulfate metabolism (Spearman, $P = 0.017$; Fig. 2k) and between *Deltaproteobacterial hgcA* copies and soil Hg methylation ($P < 0.01$; Fig. 2l) or sulfate content ($P = 0.004$; Supplementary Fig. 8), suggesting again that SRB are the primary drivers of FQ-induced MeHg increases in paddy soils. We further performed clade-specific *hgcA* gene profiling targeting *Deltaproteobacteria* and Archaea, which primarily encompass *hgcA*-SRB, *hgcA*-FeRB, and *hgcA*-methanogenic Archaea. Both *Deltaproteobacterial-hgcA* and Archaeal-*hgcA* sequencing analyses showed that ENR treatment did not significantly alter the overall α-diversity or community composition of the *hgcA*-containing *Deltaproteobacteria* and Archaea (Supplementary Figs. 9–11, Tables 6–8; Supplementary Text 2). Although ENR treatment caused minor changes in the relative abundance of the *hgcA*-SRB (Supplementary Fig. 10a, b), a significant correlation between the relative

https://doi.org/10.1038/s41467-026-79083-7

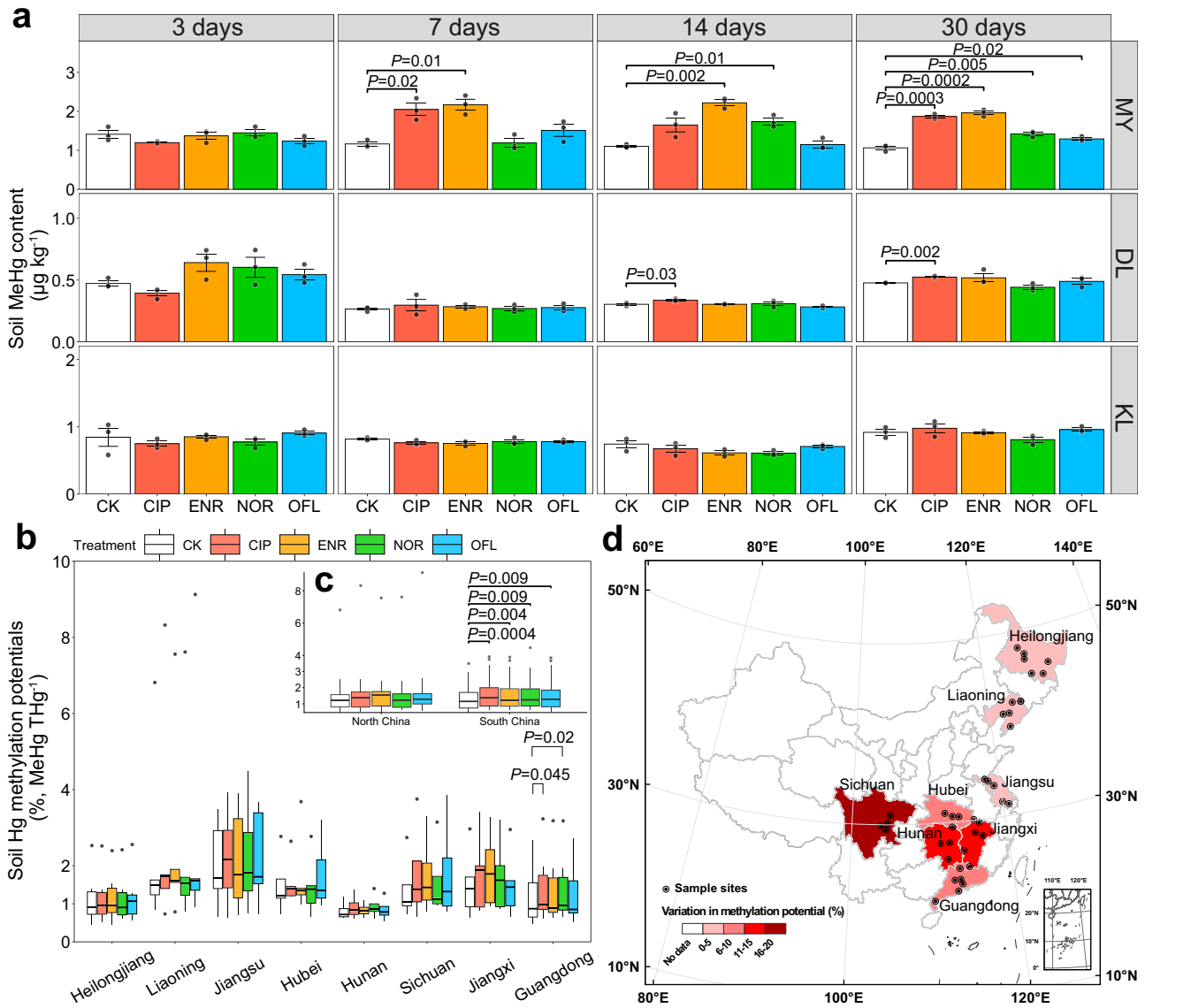


Fig. 1 | Stimulatory effects of fluoroquinolone (FQ) antibiotics on methylmercury (MeHg) production in rice soils. a MeHg in three soil types dominated by different Hg-methylating microbes: sulfate-reducing bacteria (SRB; MY), iron-reducing bacteria (FeRB; DL), and methanogens (KL), following FQ treatment over varied incubation periods; data are presented as mean ± standard deviation ($n = 3$), and statistical significance was assessed using a two-tailed unpaired *t*test. **b** Soil Hg methylation potentials in paddy soils from eight provinces across China. **c** Soil Hg methylation potentials in pooled soil samples from North and South China. **d** Overall percentage increases in soil MeHg contents after FQ treatment; each point represents a sampling site, and the base map of China is provided by the Resource and Environment Science and Data Center (https://www.resdc.cn/Datalist1.aspx?FieldTyepID=20,0, accessed on 18 October 2022). Treatments included a CK fluoroquinolone-free control, CIP ciprofloxacin, ENR enrofloxacin, NOR norfloxacin, and OFL ofloxacin. In (**b**) and (**c**), the center line represents the median; the lower and upper quartiles of the box represent the interquartile range (IQR, 25–75%); whiskers extend to 1.5 times the IQR; vertical lines represent the standard deviation (SD) values; black dots represent outliers; and statistical analysis was performed using a two-tailed paired *t*-test.

hgcA-SRB abundance and soil MeHg content was indicated ($P = 0.007$; Supplementary Fig. 10c). In contrast, the relative abundance of *hgcA*-containing *Geobacter* spp. did not significantly correlate with soil MeHg content (Supplementary Fig. 11). Meanwhile, iron reduction was only marginally and inconsistently affected in the +Mo group, ruling out a major contribution from *hgcA*-FeRB, another key group of Hg-methylating microbes[27], to the enhanced MeHg synthesis (Supplementary Fig. 12). Together, these findings imply that *hgcA*-FeRB and *hgcA*-Methanogens are unlikely to be key drivers underlying the FQ-stimulated MeHg synthesis.

To assess whether FQ stress broadly shapes the overall microbial community or specifically targets *hgcA*-harboring SRB in natural soils, 16S rRNA gene profile analysis in ENR-spiked soils was performed to link microbial responses under FQ exposure to MeHg production. ENR was found not to significantly alter microbial α-diversity (as measured by Shannon index, Chao_1, and OTU counts; Supplementary Fig. 13). For the microbial composition, a total of 11 phyla were identified in the eight tested soils (Fig. 3a). Non-metric multidimensional scaling (NMDS) based on Bray–Curtis dissimilarities of environmental microbiome showed the separation of the microbial community composition under ENR treatment (stress = 2.6e-06; Fig. 3b). PERMANOVA analysis, stratified by soil origin, showed that the ENR treatment had a marginal separation effect on the microbial community composition (pseudo-$F_{1,6} = 1.452$, $R^2 = 0.195$, $P = 0.061$). Among the microbial communities, *Firmicutes* and *Proteobacteria* were shown to be the top two dominant phyla, comprising 21.4–47.1% and 16.5–28.7%, respectively

https://doi.org/10.1038/s41467-026-79083-7

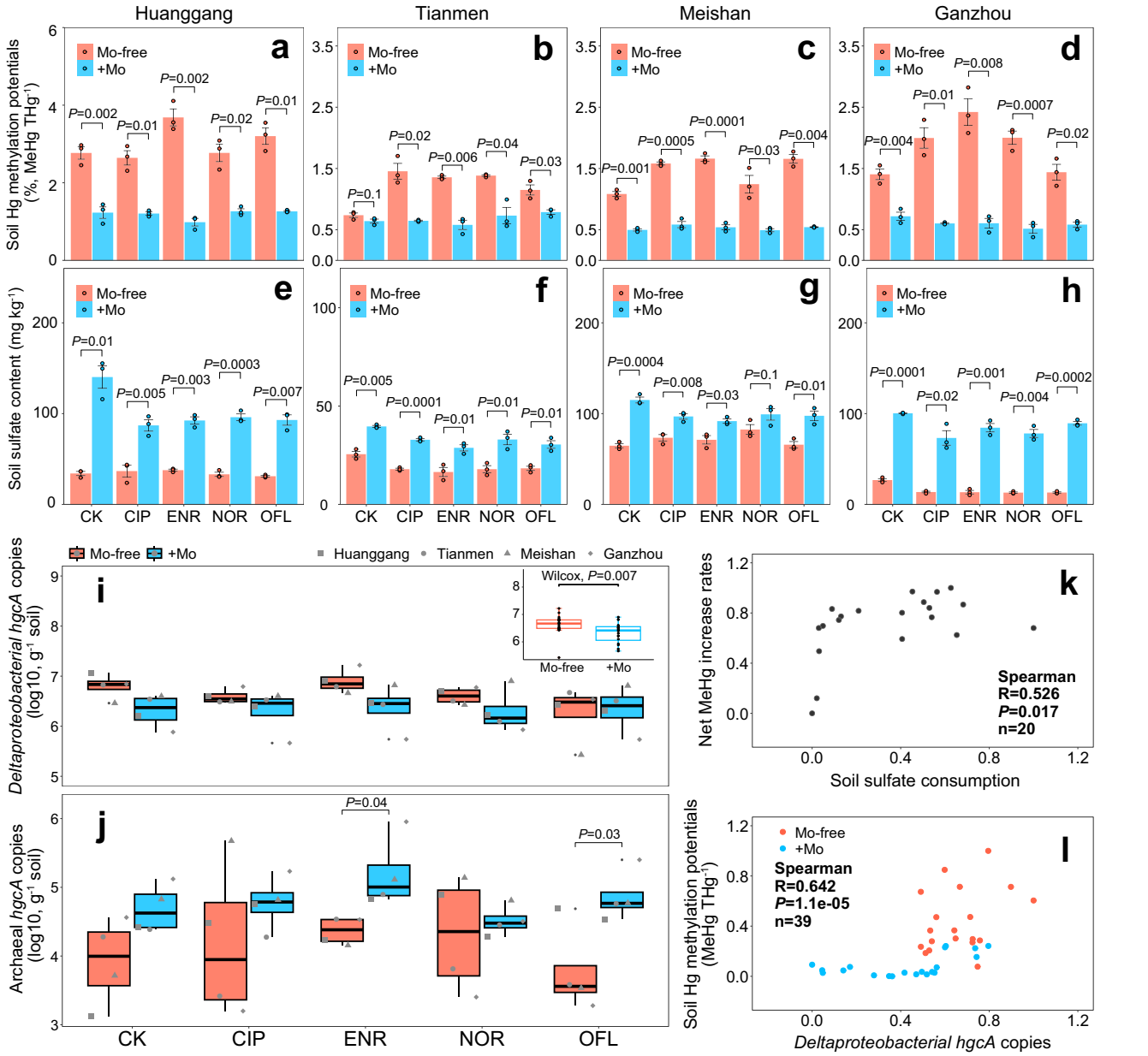


Fig. 2 | Soil mercury (Hg) Methylation and sulfate levels, copies of the Hg-methylating gene *hgcA*, and their correlations in fluoroquinolones (FQ)-amended soils under the impact of the sulfate reduction inhibitor molybdate. **a–d** Soil Hg methylation potentials in the FQ-amended soils spiked with molybdate (+Mo) or in Mo-free soil controls (Mo-free). **e–h** Dissolved sulfate concentrations in the FQ-amended soils with +Mo and Mo-free treatments. **i** Copy numbers of *hgcA* from *Deltaproteobacteria* under the same conditions. **j** Copy numbers of *hgcA* from Archaea under the same conditions. **k** Correlation between sulfate consumption and net MeHg increase rates. **l** Correlation between *Deltaproteobacterial hgcA* gene abundance and soil Hg methylation potentials (Noted: upon examination of the low values, we identified one value as an outlier. The outlier was excluded from the analysis, and its removal did not affect the statistical significance of the observed relationship). Treatments include a fluoroquinolone-free control (CK, Mo-free), ciprofloxacin (CIP), enrofloxacin (ENR), norfloxacin (NOR), and ofloxacin (OFL). In (**a–h**), data are presented as mean ± standard deviation ($n = 3$). In (**a–h**,**i**,**j**), statistical analysis was performed using a two-tailed unpaired *t*-test. In (**i**,**j**), the center line represents the median; the lower and upper quartiles of the box represent the interquartile range (IQR, 25–75%); whiskers extend to 1.5 times the IQR; vertical lines represent the standard deviation (SD) values; black dots represent outliers; and different point symbols represent individual soil samples. Both *Deltaproteobacterial hgcA* and Archaea *hgcA* were amplified from genomic DNA. In (**k**,**l**), Spearman's rank correlation with a two-sided test.

(Fig. 3c). In general, ENR amendments increased the relative abundance of *Firmicutes*, while decreasing that of *Proteobacteria* in comparison with the control (CK) soils (Fig. 3c–e). As to the microbial composition, Venn diagram analyses further showed that a total of 502 OTUs of *Proteobacteria* (Fig. 3d) and 256 OTUs of *Firmicutes* (Fig. 3e) were shared by the four soils. ENR addition caused a species reduction in *Proteobacteria* and *Firmicutes* by 36.7–51.3% and 27.1–40.5%, respectively, indicating both phyla show notable responses to FQ stress. Furthermore, 16S rRNA gene-based functional inference (PICRUSt2) implies that FQ input likely modulates microbial activities (Supplementary Fig. 14, details in Supplementary Text 3). Considering that most completely oxidizing SRB metabolize organics through the reductive acetyl-CoA pathway[15], we integrated functional predictions with taxonomic profiles and identified four SRB possessing this pathway in the soils as putative Hg methylators, i.e., *Syntrophobacter*[21], *Desulfuromonas*[28], *Desulosporosinus*[15], and *Desulfitobacterium*[15] (Supplementary Figs. 14–16). ENR treatment significantly increased the relative abundance of these SRB groups, which was positively

https://doi.org/10.1038/s41467-026-79083-7

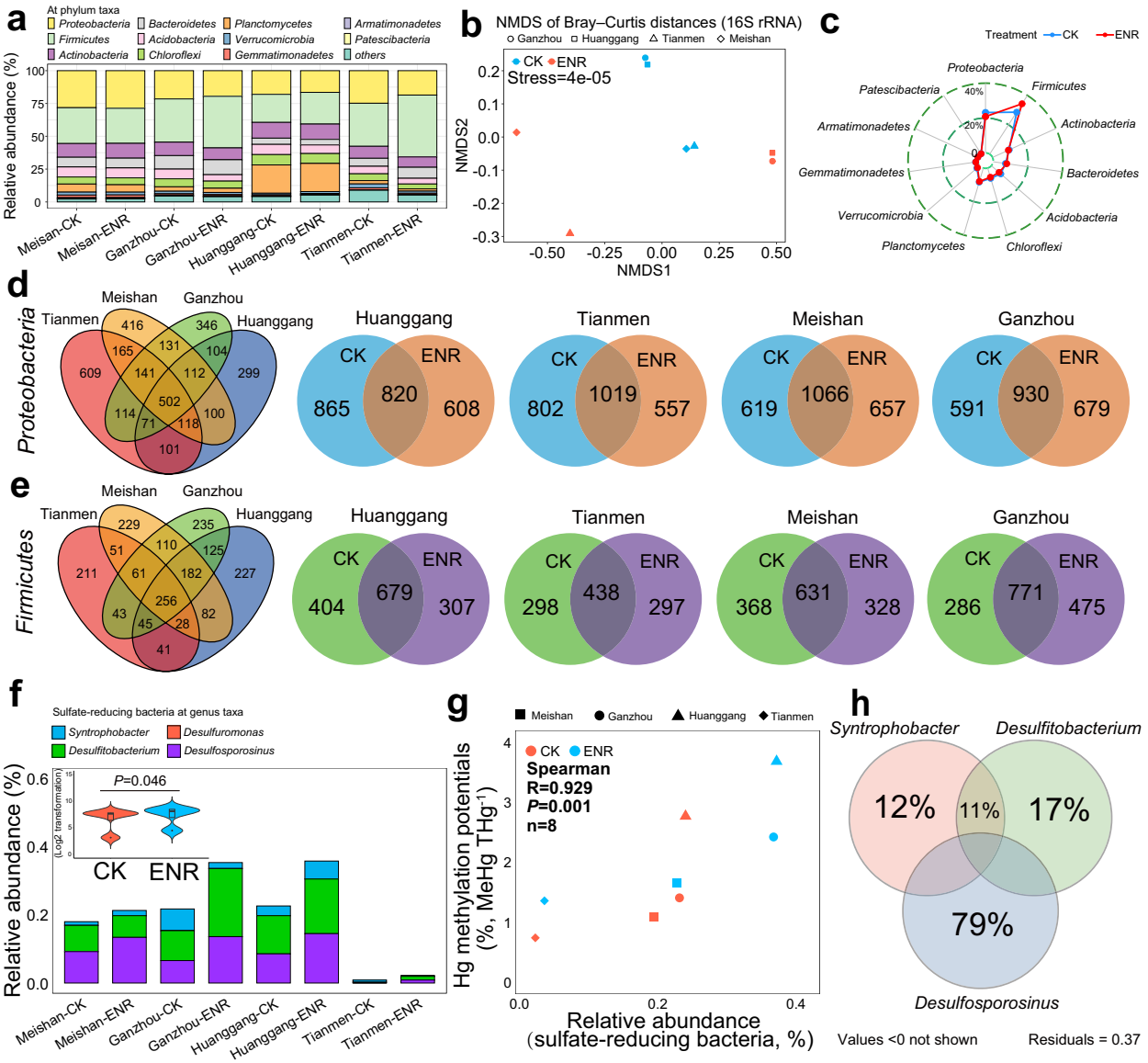


Fig. 3 | Effects of enrofloxacin (ENR) amendment on soil microbial community characterized by 16S rRNA gene sequencing. a Changes in soil microbial community composition across different sampling sites. **b** Non-metric multi-dimensional scaling (NMDS) analysis based on Bray–Curtis dissimilarities of 16S rRNA gene profiles. **c** Effects of ENR on the relative abundance of microbial phyla. **d**,**e** Operational taxonomic unit (OTU) overlaps and distinctions in *Proteobacteria* (**d**) and *Firmicutes* (**e**) among varied soils under ENR stress, as shown by Venn diagram analyses. **f** Relative abundance of sulfate-reducing bacteria (SRB) as potential Hg methylators, in the violin-box plot, the center line represents the median; the lower and upper quartiles of the box represent the interquartile range (IQR, 25–75%); whiskers extend to 1.5 times the interquartile range (IQR); black dots represent outliers; statistical analysis was performed using a two-tailed paired *t*-test. **g** Correlation between the relative abundance of SRB and Hg methylation potentials, and Spearman's rank correlation with a two-sided test. **h** Variance partitioning analyses showing the potential contributions of SRB groups to MeHg content in soils.

correlated with soil Hg methylation potentials ($P = 0.001$; Fig. 3f, g). Among the dominant Hg methylators representing 63% of changes in MeHg content, the primary contributors included *Desulfitobacterium* and *Desulfosporosinus* from *Firmicutes* and *Syntrophobacter* from *Proteobacteria* (Fig. 3h). Consistent patterns observed from 16S rRNA gene analysis and *Deltaproteobacterial-hgcA* gene profiling indicate an association between SRB and FQ-enhanced MeHg production.

The key role of sulfate reduction and SRB activity in driving FQ-induced MeHg production was further supported by Partial Least Squares Path Modeling (PLS-PM) analyses, which integrated the overall geochemical and molecular biological data presented above (GoF of 0.731–0.665; Fig. 4). In FQ-contaminated soils, both abiotic factors (e.g., dissolved sulfate) and microbial factors (particularly *hgcA*-SRB) contributed to MeHg synthesis. Although dissolved sulfate has often been reported to exert only a weak influence on SRB-mediated MeHg production[29,30], likely because it is utilized by both methylating and non-methylating SRB, FQs increased the *hgcA*-SRB/SRB ratio, thereby enhancing MeHg production via sulfate-dependent pathways (total effect = 0.425; Fig. 4a). According to the model (Fig. 4b), the principal biotic contributors to Hg methylation were putative *hgcA*-carrying SRB (total effect = 1.30), while *hgcA*-containing SRB were affiliated with *Deltaproteobacteria* (total effect = 1.55) or other microbial clades. These microorganisms are widespread in soils and sediments and play a significant role in MeHg synthesis[15,18,31]. Based on the presented modeling results, we infer that multiple, non-mutually exclusive processes may contribute to the elevated Hg methylation potential

https://doi.org/10.1038/s41467-026-79083-7

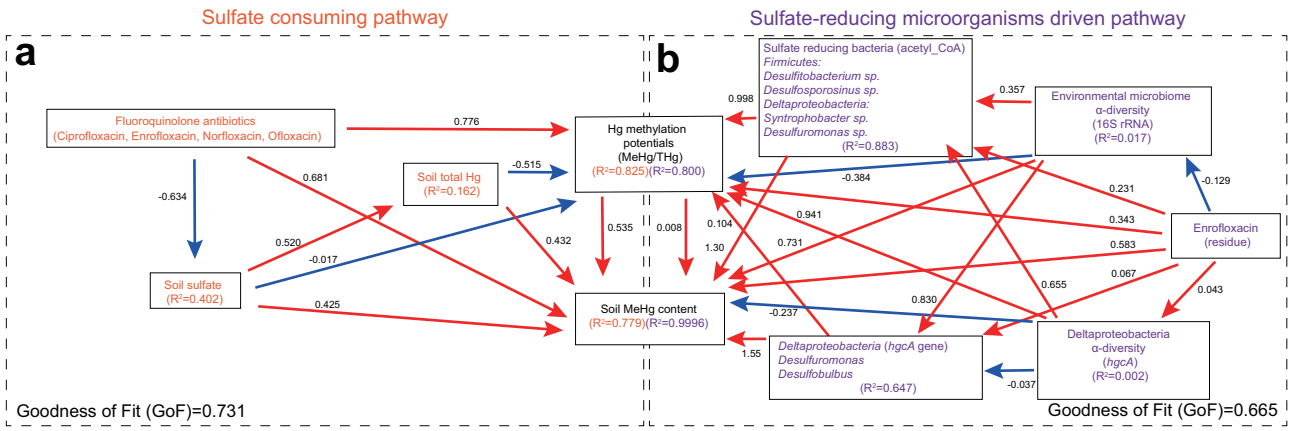


Fig. 4 | Multiscale pathways illustrating how fluoroquinolone antibiotics stimulate methylmercury (MeHg) production in paddy soils. a Partial Least Squares Path Modeling (PLS-PM) depicting a sulfate consumption-mediated pathway in which fluoroquinolone exposure enhances MeHg synthesis by regulating sulfate consumption. **b** PLS-PM predicting an SRB-driven pathway whereby enrofloxacin stimulates MeHg production by enhancing the activity of SRB.

observed in FQ-amended soils. These specifically include: 1) microbiome restructuring, reflected by shifts in the abundance of Hg-methylating bacteria under FQ exposure; 2) selection and stimulation of SRB, particularly *hgcA*-SRB, as indicated by changes in the *hgcA*-SRB/total-SRB ratio; 3) enhanced metabolic activity in SRB, potentially linked to carbon and one-carbon metabolism associated with the acetyl-CoA pathway. Notably, the biochemical linkage between acetyl-CoA metabolism and *hgcA* function should be interpreted as a functional correlation rather than a direct mechanistic pathway and thus warrants further investigation, given that this association remains unresolved[17,18].

FQ stimulates MeHg synthesis in enriched SRB cultures

To further elucidate the mechanism underlying elevated Hg methylation by SRB under FQ stress, we enriched SRB from the four tested soils in Batch assay III using an SRB enrichment medium (Supplementary Table 9). The resulting consortia were then subjected to FQ exposure assays, Hg methylation quantification, and metagenomic sequencing, as summarized in Supplementary Fig. 17 (Batch assay IV).

We first evaluated the tolerance to FQ stress of the SRB consortia enriched separately from the four studied soils using four different carbon resources (i.e., acetate, lactate, propionate, and butyrate). After 120 h of incubation, none of the consortia were able to grow in media amended with ENR, CIP, or OFL (OD_{600}; Fig. 5a–d), whereas lactate-enriched SRB could grow in the presence of NOR. To further explore microbiome tolerance responses, lactate-enriched SRB consortia were selected for subsequent assays. High-density cultures ($OD_{600} > 1$) were exposed to ENR for 8 h (Fig. 5e) and the overall consortium growth was not significantly inhibited by ENR during the subsequent 24 h incubation (Fig. 5e). Once the ENR stress disappeared (ENR-rec; where enriched consortia were first centrifuged after 8 h of ENR exposure to remove residual antibiotic and then resuspended in antibiotic-free medium), only the consortia from Tianmen and Meishan soil could grow in the antibiotic-free medium (Fig. 5e). These results indicate a transient tolerance of enriched consortia at the community level.

We further investigated how Hg methylation responds to ENR stress using lactate-enriched SRB consortia. The SRB consortia were exposed to inorganic Hg solution and incubated for 24 h. ENR exposure significantly stimulated MeHg production in consortia derived from most soils, with the exception of Meishan soil (Fig. 5f). Hg methylation potentials in ENR-treated cultures ranged from 5.3% to 48.0%, compared to a range of 0.8%–13.3% in ENR-free groups (Supplementary Table 10, Supplementary Fig. 18 and Supplementary Text 5). In ENR-recovery treatments (ENR-rec), Hg methylation potentials of these recovered cultures were largely inhibited relative to the ENR treatment (Fig. 5f), except for the enriched SRB from Huanggang.

To characterize the taxonomic-level responses of SRB and *hgcA*-containing communities to FQ stress, we performed metagenomic analyses on enriched consortia exposed to ENR for 8 hours or left unexposed. Analysis of *dsrA* or *dsrB* genes indicated that SRB accounted for 43.9%–82.4% of the total microbial community in enrichments derived from the four studied soils (Supplementary Table 11). FQ exposure altered the relative abundance of individual taxa within these consortia, with specific effects on SRB lineages (Supplementary Fig. 19). When data from all SRB enrichments were combined, ENR exposure significantly inhibited the overall growth of SRB consortia ($P = 0.003$; Fig. 5g(**i**)). Specifically, ENR exposure did not suppress the growth of complete-oxidizing SRB groups carrying the reductive acetyl-CoA pathway ($P = 0.93$; Fig. 5g(**ii**)) but significantly inhibited the growth of SRB groups with other carbon metabolism pathways in the consortia ($P = 0.0006$; Fig. 5g(**iii**)), suggesting SRB with a reductive acetyl-CoA or complete SRB may be less sensitive to ENR exposure under the tested conditions. Moreover, metagenomic analysis identified 14 genes associated with the acetyl-CoA pathway, among which the abundances of *folD*, *fdhB*, and *fhs* were enriched under ENR stress (Supplementary Fig. 20a), indicating sustained microbial activity in tolerant SRB. Analysis of *hgcA*-containing communities revealed 10 genera, all within *Firmicutes* (Supplementary Fig. 21). NMDS based on Bray–Curtis dissimilarities at the genus level showed that soil type was the primary driver shaping *hgcA*-containing community composition, rather than antibiotic exposure (Stress = 5.8e-05; Fig. 5h). PERMANOVA analysis further revealed no significant compositional response of Hg-methylating microbial communities to ENR stress (pseudo-$F_{1,6} = 0.12$, $R^2 = 0.02$, $P = 0.86$). The total abundance of the *hgcA* gene showed a consistent non-significant response to ENR exposure ($P = 0.62$, Fig. 5i). Importantly, quinolone resistance genes *lfrA* and *norB* were detected in enriched SRB consortia from Huanggang, Ganzhou, and Meishan soils, but were absent in Tianmen soil (Supplementary Table 12). This indicates that the presence of these resistance genes is not necessarily responsible for the enhanced MeHg production under ENR stress, implying that antibiotic resistance traits are not directly coupled to Hg methylation capacity in SRB consortia.

Discussion

In this study, we demonstrated that FQs promote MeHg accumulation in complex soil systems at environmentally relevant concentrations (i.e., 10 μg kg^{-1} and 10 mg kg^{-1}), although such increases were not

https://doi.org/10.1038/s41467-026-79083-7

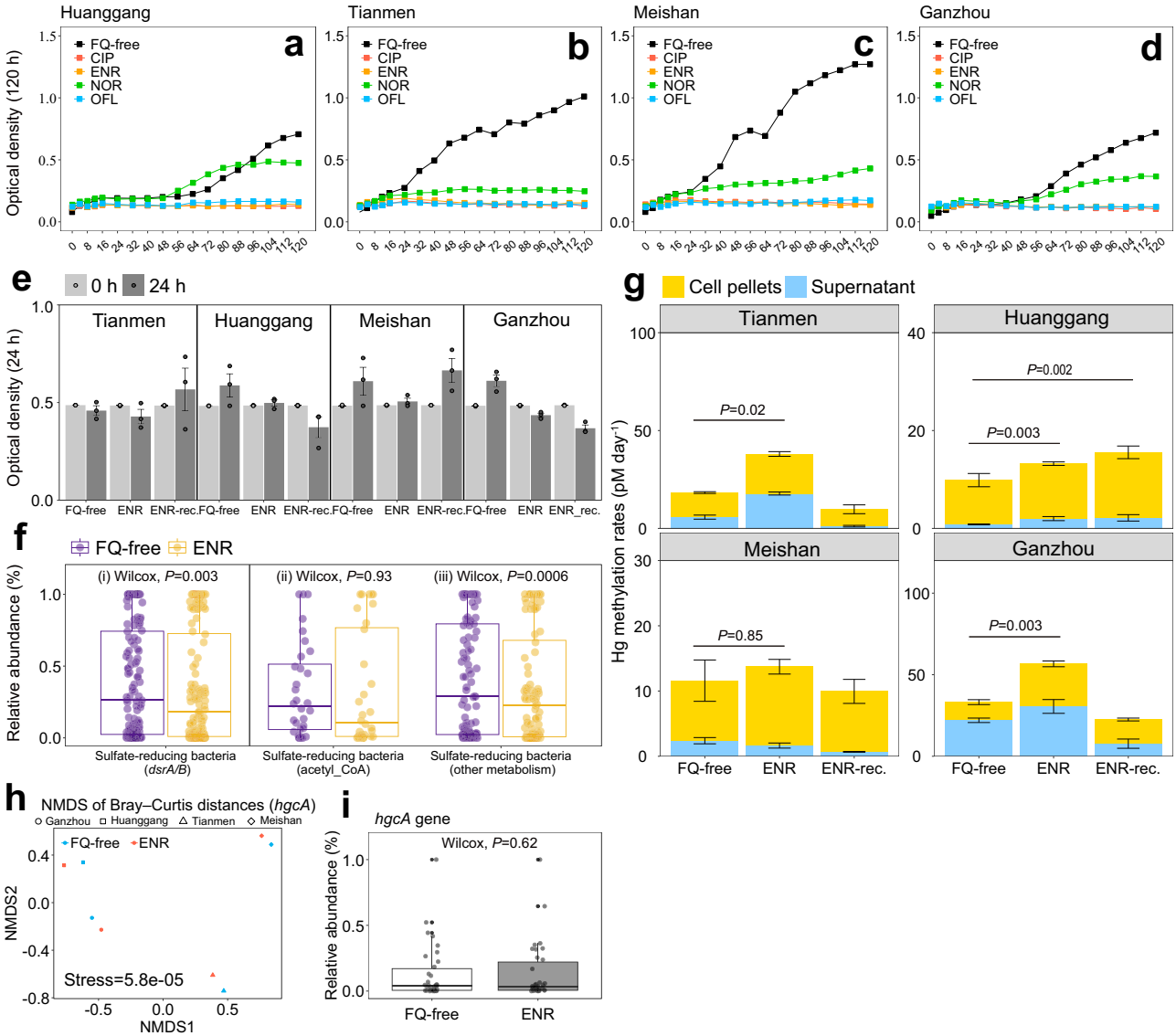


Fig. 5 | Effects of fluoroquinolones (FQs), particularly enrofloxacin (ENR), on the growth, community composition, and mercury methylation potentials of enriched sulfate-reducing bacteria (SRB). a–d Growth dynamics over 120 hours of SRB consortia enriched from four different soil sampling sites, each exposed to one of four FQs (10 mg L^{-1}). **e** Comparison of SRB growth with lactate as a carbon source at 24 h under ENR exposure (ENR) and after ENR removal (ENR-rec), for consortia from the four sampling sites, and data are presented as mean ± standard deviation ($n = 3$). **f** Hg methylation rates of SRB consortia from (**e**) comparing ENR-treated and ENR-rec cultures at 24 h, data are presented as mean ± standard deviation ($n = 3$), and statistical analysis was performed using a two-tailed paired *t*-test. **g** Relative abundance (%) of total SRB, SRB harboring the acetyl-CoA pathway, and SRB with other carbon metabolism pathways, based on metagenomic analyses of dissimilatory sulfite reductase genes (*dsrA/B*) and other key pathway genes across the four combined cultures. **h** Non-metric multidimensional scaling (NMDS) analyses based on Bray–Curtis dissimilarities of the *hgcA*-containing microbial communities (based on metagenomic data). **i** Abundance of *hgcA* genes based on metagenomics. Treatments included a fluoroquinolone-free control (FQ-free), ciprofloxacin (CIP), enrofloxacin (ENR), norfloxacin (NOR), and ofloxacin (OFL). In (**g**) and (**i**), the center line represents the median; the lower and upper quartiles of the box represent the interquartile range (IQR, 25–75%); whiskers extend to 1.5 times the IQR; vertical lines represent the standard deviation (SD) values; the point symbols represent individual microbial relative abundance or abundance of microorganism, and all data were normalized 0–1 scale before plotting. In (**f**) and (**i**), statistical analysis was performed using a paired Wilcoxon signed-rank test.

observed in all soils or across the entire incubation period. This effect is primarily driven by enhanced Hg methylation rather than reduced MeHg demethylation (Supplementary Fig. 7), although soil MeHg levels ultimately reflect the net balance between these two processes[32,33]. In natural ecosystems, Hg methylation is governed by both abiotic factors, such as pH and inorganic Hg availability, and biotic factors, particularly the abundance and activity of Hg-methylating microorganisms[34]. Given that FQ treatments exerted minimal effects on pH, total soil Hg content, and dissolved Hg concentrations under experimental conditions (Supplementary Figs. 3–4), changes in abiotic factors were unlikely to explain the FQ-enhanced MeHg production. We therefore hypothesize that FQs increase soil MeHg levels predominantly through microbial mechanisms. This inference is further supported by metabolism inhibition experiments using molybdate, in which the suppressed MeHg production was consistent with the inhibition of sulfate reduction (Fig. 2).

Among the diverse Hg-methylating microorganisms, *hgcA*-SRB were identified as the dominant microbial methylators driving elevated Hg methylation under FQ exposure, with *hgcA*-FeRB and *hgcA*-Methanogens playing minor roles. Integrative analyses linking Hg methylation potentials (Fig. 2a–d), sulfate-reduction inhibition (Fig. 2e–h), dissolved sulfate concentrations (Fig. 2k and

Supplementary Fig. 8), and *Deltaproteobacteria*-associated *hgcA* gene abundance (Fig. 2l) collectively suggest that FQs enhance MeHg synthesis in paddy soil by modulating the metabolic activity of *hgcA*-SRB. Although FeRB were identified as a key predictor of MeHg production in paddy soils[29] and shown to drive MeHg formation in the presence of CIP in a pure-culture system[23], they were not the primary contributors to the observed increases in Hg methylation in this study. This conclusion is supported by the absence of a significant correlation between the relative abundance of *hgcA*-containing *Geobacter* spp. and soil MeHg content (Supplementary Fig. 11). Meanwhile, FeRB, regardless of the presence of Hg methylating genes, can also participate in Hg methylation, likely by regulating inorganic Hg (IHg) mobility through iron reduction[35]. However, the concentration of dissolved IHg remained unchanged after FQ addition (Supplementary Fig. 4), further underscoring a minor role for FeRB in this process. The contribution of *hgcA*-Methanogens is also likely negligible, as ENR exposure did not significantly alter the diversity or composition of archaeal methanogens. Moreover, a large-scale study revealed that the relative abundances of methanogens were negatively correlated with Hg methylation potential in paddy soils, suggesting that these organisms may play a more important role in MeHg degradation[27].

The observed response of *hgcA*-SRB to FQ exposure is mainly driven by the enhanced microbial activity, particularly among those utilizing the acetyl-CoA pathway, rather than by changes in their abundance. Both the results of 16S rRNA of the soil microbiome (Fig. 3f) and metagenomic analyses based on the enriched consortia (Fig. 5g) showed that SRBs with the reductive acetyl-CoA pathway were less sensitive to FQs. Accordingly, genes associated with this pathway, including *fhs* and *folD*, displayed significant increases in relative abundances under ENR treatment ($P < 0.05$; Supplementary Fig. 20b). Both *fhS* and *folD* are key components of tetrahydrofolate-mediated one-carbon metabolism and ATP generation[36,37], suggesting that this group of SRB continues to maintain these activities under ENR exposure. These findings point to a potential mechanism whereby ENR-tolerant SRB enhance their carbon metabolism capacity under antibiotic stress and consequently their ability to compete for available nutrients. In contrast, antibiotic-sensitive microorganisms may be inhibited or eliminated by FQ, indirectly increasing the relative abundance of tolerant SRB in soil, consistent with an r-strategy-like ecological response[38]. A subset of these tolerant SRB harbored the *hgcAB* genes, thereby contributing to MeHg production. Notably, both the overall abundance and community composition of *hgcA*-containing microorganisms remain stable under ENR exposure, suggesting that stimulatory effects of FQs on Hg methylation were unlikely driven by shifts in methylator abundance or community restructuring. Instead, FQ-enhanced MeHg production may primarily rely on increased metabolic activity of Hg-methylating microorganisms, which can be further validated based on transcript analyses. In line with this, metagenome-assembled genomes of *hgcA*-harboring *Desulfosporosinus* revealed that changes in *hgcA* abundance at the population level were positively correlated with MeHg production rates, rather than the total number of *hgcA* copies in the community (Supplementary Text 4; Supplementary Fig. 22).

In agricultural ecosystems, FQ contamination could disrupt indigenous microbial communities in paddy soils (Fig. 3a–c). Their primary mode of action involves inhibition of DNA gyrase in Gram-negative bacteria and topoisomerase IV in Gram$^+$ bacteria[39]. Because FQs predominantly target aerobic organisms, their activity against anaerobic bacteria is generally weaker and more variable, with CIP and OFL typically showing limited effectiveness[40,41]. However, detailed information on their specific effects remains scarce[42]. In the paddy soils studied here, FQ addition might inhibit FQ-sensitive soil bacteria that compete with SRB for shared resources, such as organic carbon as electron donors. Reduced competition for substrates like organic acids and hydrogen probably enabled FQ-tolerant SRB to grow and metabolize more actively than untreated controls. This occurs despite the observed growth inhibition of incomplete-oxidizing SRB strains under ENR, CIP, or OFL exposure (Fig. 5). The resultant increase in SRB activity, coupled with diminished microbial competition, enhances microbial MeHg production. Such community-level dynamics align with patterns documented in other anoxic environments, where a consortium of microbial groups is required to fully degrade organic matter[43,44]. In homogenized slurries where natural redox gradients are disrupted, SRB and other respiring bacteria tend to outcompete methanogens for electron donors like H_2[45,46].

In summary, this study demonstrates that FQ contamination promotes net MeHg production in paddy soils by stimulating the activity of Hg-methylating microorganisms, rather than altering the abundance or microbial composition of Hg methylators. Antibiotic-driven ecological filtering favors FQ-tolerant SRB populations, within which Hg-methylation potential is not only maintained but functionally amplified. Notably, further field investigations regarding the impact of FQ on net MeHg production are still warranted, particularly given the differences between constant experimental conditions and the changing field conditions (such as temperature, pH, and redox potential). In addition, the complex plant-soil interactions and the long-term impacts were not addressed in this work. These limitations implicate existing knowledge gaps regarding the impact of FQ contamination on MeHg production in paddy fields and its subsequent bioaccumulation in rice grains. Nevertheless, these findings highlight microbial physiological responses as a critical but often overlooked pathway in which antibiotic contamination enhances Hg methylation in soil ecosystems, thereby potentially promoting the entry of bioavailable, neurotoxic MeHg into human food chains.

Methods

Soils and chemicals

A total of 41 paddy soils were collected from nine major rice-planting provinces in China from 2018 to 2019. Soils were collected at a depth of 0–20 cm and stored at 4 °C prior to lab experiments. The detailed description of soil characteristics is shown in Supplementary Table 2. FQs used in this study included Ciprofloxacin (CIP, Macklin, 98%), Enrofloxacin (ENR, Macklin, 98%), Norfloxacin (NOR, Macklin, 98%), and Ofloxacin (OFL, Macklin, 98%). The stock solutions of FQs were prepared with sterile water at 121 °C for 20 min (Zealway) and stored at 4 °C.

Soil incubation experiments

Batch assay I-Hg methylating microbes. Three soils with known composition of microbial Hg methylators were selected for the assays: MY soil (with a relatively high abundance of *hgcA*-containing SRB, 31.5%), DL soil (with a relatively high abundance of *hgcA*-containing FeRB, 74.4%), and KL soil (with a relatively high abundance of *hgcA*-containing methanogens, 39.3%). Detailed information on the microbial methylator composition of these three soils can be found in Supplementary Table 1. To be specific, 2.0 g of soil was weighed and placed in a 50 mL-sterilized centrifuge tube, and then 2 mL of oxygen-removed sterile water (for the control CK) or 2 mL of the corresponding FQ solutions (for CIP, ENR, NOR, or OFL-treated soils) was separately added to reach a final concentration of 10 μg kg^{-1} (here only, representing a low FQ contamination scenario)[47] and 10 mg kg^{-1} (here and in subsequent experiments, reflecting a heavy FQ contamination). A FQ concentration of 10 mg kg^{-1} was used here and in subsequent experiments, given that comparable concentrations (3.06–9.8 mg kg^{-1}) of CIP, NOR, and ENR were detected in agricultural soils[11,48]. Three replicate tubes were set for each treatment, and all these steps were conducted in an anaerobic glove box (95% N_2 and 5% H_2). After thorough mixing, these tubes were incubated in the dark at 25 °C. On days 3, 7, 14, and 30, three tubes from each treatment were randomly selected and lyophilized for soil MeHg analyses. The supernatants

were analyzed for pH (HQ40d, HACH), soil THg, and dissolved THg contents (Meishan only; Supplementary Method 1).

Batch assay II—National survey. Forty soils, with two soils from Batch assay I (MY and DL) and 38 additional soils from eight provinces in China (Supplementary Table 2), were used to investigate the effects of four FQs (CIP, ENR, NOR, and OFL) on net MeHg production. The incubation was carried out as previously described. After 21 days of incubation, soils were lyophilized and stored at -20 °C for the analyses of MeHg (details below), FQ residue contents, and soil THg contents (details below).

Batch assay III-metabolism assay. Based on results from Batch assay II, the four soils showing the highest MeHg increases following FQ addition, i.e., Huanggang, Tianmen, Meishan, and Ganzhou, were selected for metabolism study. In the metabolism assay, four FQs were added as described in Batch assay I. The data obtained in the batch assays without adding molybdate were grouped as Mo-free. Molybdate (in the form of Na_2MoO_4, Aladdin, > 98%) was added at a final concentration of 20 mM, based on the sulfate levels in the native paddy soils[49], to achieve complete inhibition of sulfate reduction. The effects of molybdate doses, i.e., 5, 10, and 20 mM, were evaluated and found to be insignificant (Supplementary Text 1; Supplementary Fig. 6). These treatments are defined as the +Mo group. For samples from both Mo-free and +Mo groups, soils and supernatant were separated by centrifugation at the end of incubation. The soils were freeze-dried for the analyses of MeHg and soil THg contents (detailed below), extractable sulfate[50] (extracted using deionized water at a soil-to-water ratio of 1:5 and then analyzed using ion chromatography, IC-900, Thermos Fisher), *hgcA* copies (detailed below), microbial composition of *hgcA*-containing microorganisms (Supplementary Method 2), 16S rRNA gene sequencing (detailed below), FQ residues (detailed below), and Fe reduction (Supplementary Method 3). Another parallel incubation was conducted as described above to quantify the rate constants of Hg methylation and MeHg demethylation using a stable isotopic Hg-tracing technique (Supplementary Method 4). Data obtained from Batch assay III were also subjected to the PLS-PM analyses (details below).

Batch assay IV-SRB enrichment. To investigate how FQ stress affects SRB and Hg methylation, SRB were first enriched with an SRB medium supplemented with one of four representative carbon sources to explore their survival strategies and identify potential drivers of elevated Hg methylation (Fig. 5a). To be specific, 1.0 g of soil (Huanggang, Tianmen, Meishan, or Ganzhou) was added to a sterile serum bottle containing 10 mL of double-distilled water in an anaerobic glove box (95% N_2 and 5% H_2). The bottles were then sealed and incubated at 30 °C in the dark to revive SRB growth for seven days[51]. Then 1 mL of the soil slurry was pipetted and injected into a serum bottle containing 40 mL of SRB medium based on the components listed in Supplementary Table 9[52]. All containers, tools, H_2O, and liquid media were sterilized at 121 °C for 20 min prior to use to prevent contamination. The manipulation processes were strictly conducted in an anaerobic bag containing 95% N_2 and 5% H_2.

To determine whether SRB survive FQ exposure through resistance or tolerance, four previously reported carbon sources, i.e., acetate, lactate, propionate, and butyrate, were added separately to each bottle at a final concentration of 20 nM. The microcosms containing a carbon source, soil slurry with SRB, and liquid media were incubated at 30 °C in the dark. When the media turned black, which indicated high SRB activity (Supplementary Fig. 23), 1 mL of the cell suspension was transferred into a new serum bottle containing the same carbon source and 40 mL of liquid medium. For each carbon source, five treatments were established: a control (FQ-free) without any FQ, and four with individual FQs—CIP, ENR, NOR, and OFL. Each treatment had three replicates. Each of the FQ stock solutions was added to a final concentration of 10 mg L^{-1}. The bottles were then incubated again at 30 °C in the dark. Optical density was measured (HACH, DR1900) every four hours during the first 24 h, then every eight hours thereafter until 120 h (Fig. 5).

To evaluate the Hg methylation potential of the enriched microorganisms, an additional cultivation experiment was conducted. Lactate was used as a carbon source, as it supported better SRB growth (Supplementary Fig. 23). Once SRB were activated, 1 mL of cell suspension was transferred into a medium containing 10 mg L^{-1} ENR and incubated for 8 h. A control treatment without ENR addition was included for comparison (FQ-free). After incubation, 10 mL of cell suspension with (ENR treatment; Fig. 5e, g) or without ENR (FQ-free; Fig. 5e, g) was transferred into a sterilized tube, followed by the addition of 50 µL of 1 mg L^{-1} $Hg(NO_3)_2$, resulting in a final Hg concentration of 25 nM. One 10 mL culture was centrifuged and resuspended in blank SRB medium to test the growth recovery and Hg methylation (ENR-rec; Fig. 5e, g). The mixtures were thoroughly stirred and incubated in the dark at 30 °C for 24 h. At the end of incubation, cell suspensions were divided into four portions: The first fraction was used to measure optical density at 600 nm (OD_{600}). The second was centrifuged and used for MeHg analysis (details below). The third fraction (1 mL) was transferred into fresh ENR-free medium for further cultivation and subsequent quantification of Hg methylation potentials as described above. The fourth fraction was used for metagenome sequencing to investigate the mechanisms underlying enhanced Hg methylation by SRB under FQ stress (details below). Notably, Hg methylation potentials were measured twice in different batches (Supplementary Text 5).

Additionally, as the site with the highest relative abundance of *hgcA*-containing SRB and the greatest increase in soil MeHg contents, Mianyang soil was chosen for the enrichment and investigation of SRB in the culture medium. Briefly, 1 g of soil was suspended in 10 mL of sterilized water. After thorough mixing, 1 mL of soil suspension was transferred into the SRB enrichment medium. Four FQs were added separately to the culture bottles to achieve final concentrations of 1 µg L^{-1}, 100 µg L^{-1}, and 10 mg L^{-1}. Each treatment included four replicate bottles. For each treatment, a negative control (with sterilized water added instead of FQ solution) was also prepared with four replicates (Supplementary Fig. 24).

Parameter analyses

MeHg content. Soil samples were extracted with a 25% KOH-methanol (w/w) solution and analyzed based on the EPA 1630 method using an automated mercury Analyzer (model III, Brooks Rand)[53,54]. Briefly, 2 mL of KOH-methanol was added to a 15-mL centrifuge tube containing ~0.2 g of soil and shaken at 60 °C for 4 h. After extraction, the solution was diluted with deionized water, then filtered with a 0.22 µm filter, and then stored at -20 °C for further measurement. Standard reference materials (ERM-CC58 and TORT-3) were included for each analytical batch, and the recovery rates ranged from 80% to 120%. For analyses of MeHg in cells, 1 mL of cell suspension was centrifuged to separate the cell pellets and supernatant. The former fraction was digested using KOH-methanol as described above, and the supernatant was digested using 10 mL of 18% KBr dissolved in 5% H_2SO_4[55]. The recoveries of the matrix spike during MeHg measurements using $CuSO_4$/KBr digestion ranged from 85% to 117% (Supplementary Table 13). The method detection limit was approximately 1.78 ng L^{-1}, and the detection limit of the Hg analyzer was approximately 0.069 pg. Our analyses indicated soil MeHg contents were significantly positively correlated with THg contents across 41 soil samples in this study (Supplementary Fig. 25).

Soil total Hg content. The total Hg content in soil was determined by DMA-80 (Millstone). Standard reference materials (GSS-9) were included for each analytical batch, and the recovery rates ranged from

90.4% to 106.5%. In this study, the soil Hg methylation potentials were calculated as the ratio of MeHg to THg and expressed as a percentage.

Soil FQ content. The extraction and quantification of FQs in soils were based on a method developed by Turiel et al.[56]. Briefly, ~0.1 g of the soil spiked with enrofloxacin was weighed into a 10 mL centrifuge tube and extracted with 5 mL of 50% $Mg(NO_3)_2$ aqueous solution containing 4% aqueous ammonia. The tubes were vortexed for 10 min, then placed in an ultrasonic bath for 10 min. The samples were further centrifuged at 2900 × g for 10 min. The supernatant was purified with an HLB cartridge (CNW, 60 mg 3 mL^{-1}). This process was repeated two more times. A parallel soil sample spiked with 1 mg kg^{-1} ENR was prepared as a matrix spike prior to extraction. ENR concentrations were determined by a liquid chromatography system (Agilent, 1200 Series) at a wavelength of 280 nm and with a 20 µL injection volume. The mobile phase consisted of 85% deionized water (with triethylamine) and 15% acetonitrile (v/v), with a flow rate of 0.2 mL min^{-1}.

Sequencing and quantitative PCR of *hgcA* genes. Soil genomic DNA was isolated from 0.25 g of soil with a DNeasy PowerSoil Kit (QIAGEN) following the manufacturer's instructions. The concentration and quality of extracted DNA were determined by a NanoQuant Plate (Infinite F200 pro, TECAN). Hg methylation genes were quantified following a previous study[57]. Briefly, a 25 µL reaction mixture was prepared to contain 12.5 µL SYBR Premix Ex Taq (Takara), 2 µL of DNA template, 200 ng of bovine serum albumin (BSA), 20 µM forward primer, and 20 µM reverse primer. The *Deltaproteobacterial hgcA* genes were quantified using the primer pair ORNL-Delta-HgcA-F 5′-GCCAACTACAAGMTGASCTWC-3′ and ORNL-Delta-HgcA-R 5′-CCSGCNGCRCACCAGACRTT-3′. The amplification program included an initial denaturation at 95 °C for 3 min, followed by 40 cycles of 95 °C for 15 s, 58 °C for 15 s, and 62 °C for 15 s, and a final extension at 72 °C for 4 min. The amplification efficiency was 91.7%–102.4%. Archaeal *hgcA* genes were quantified by using the primer pair ORNL-Archaea-HgcA-F 5′-AAYTAYWCNCTSAGYTTYGAYGC-3′ and ORNL-Archaea-HgcA-R 5′-TCDGTCCCRAABGTSCCYTT-3′, by the program including an initial denaturation at 95 °C for 3 min, followed by 40 cycles of 95 °C for 15 s, 55 °C for 15 s, and 60 °C for 15 s, with a final extension at 72 °C for 4 min. The amplification efficiency was 91.9%.

For the standard plasmid and standard curve preparation, the *hgcA* genes from *Deltaproteobacteria* or Archaea were ligated to a pUCm-T Easy Vector, and an ampicillin-resistance gene was ligated to the plasmid[13,58]. Furthermore, the assembled plasmids were transformed into *Escherichia coli* JM109 cells. LB medium was prepared and supplemented with 0.1% ampicillin. The cells were inoculated into the medium and incubated at 37 °C on a shaker. Cells containing the target genes were harvested, and the target genes were extracted using the purified plasmid DNA kit (TianGen) following the manufacturer's instructions.

The PCR reaction contained 5 µL of GC buffer (5 ×), 2 µL of dNTP (2.5 mM), 1 µL of each forward and reverse primer (10 µM), 2 µL of DNA template, and 14.75 µL of ddH_2O. The thermal cycling program consisted of an initial denaturation at 98 °C for 2 min, 25 cycles of denaturation at 98 °C for 15 s, annealing at 55 °C for 30 s, and an extension at 72 °C for 30 s, followed by a final extension at 72 °C for 5 min. PCR products were purified with Vazyme VAHTSTM DNA Clean Beads (Vazyme) and quantified. After the individual quantification step, amplicons were pooled in equal amounts and subjected to paired-end 2 × 250 bp sequencing on an Illumina MiSeq platform (NovaSeq PE250) using the MiSeq Reagent Kit v3 by Shanghai Personal Biotechnology Co., Ltd (Shanghai, China). The taxonomic classification of OTUs was annotated using the Classify-Sklearn Naève Bayes Taxonomy classifier in the feature-classifier plugin against the SILVA Release 138 Database.

The community composition of *hgcA*-carrying microbes in the three representative soils (DL, MY, and KL) was amplified using the *hgcA*_261 F/*hgcA*_912 R primers with bar-code adapters according to a previous study[59]. Although these primer sets cannot cover all *hgcA*-containing taxa, their validity for environmental samples has been validated[60]. Amplicons were sequenced by the Illumina MiSeq (Illumina Inc.). Details of analyses can be found in Zhong et al.[27]

PLS-PM analyses. Partial least squares path modeling (PLS-PM) analyses were performed to elucidate the mechanism through which FQs stimulate sulfate-reducing microorganisms and subsequently enhance MeHg production in paddy soil systems based on the data from Batch assay III. This model consisted of two pathways: (i) a sulfate-consuming pathway and (ii) a sulfate-reducing microorganism-driven pathway, representing physicochemical and microbial processes, respectively. For the sulfate-consuming pathway, the results of sulfate-reduction inhibition were used to define the primary latent variables (Fig. 2). Each latent variable, including soil sulfate content, original soil THg content, soil MeHg content (MeHg/THg), and Hg methylation potential, was constructed from 8 observations (four soils with or without molybdate addition) across four FQ treatments (CIP, ENR, NOR, and OFL). For the SRB-driven pathway, the relative abundance of *hgcA*-containing SRB, α-diversity indices obtained from 16S rRNA and *hgcA* profiles, soil MeHg content, and Hg methylation potential were selected as the primary latent variables (Fig. 3 and Supplementary Figs. 6, 7, 10). Each factor was constructed using data obtained from four soils with or without ENR addition. Model robustness was assessed using bootstrapping with 999 resamples. The goodness-of-fit (GoF) index was applied to evaluate the whole model performance, with values approaching 1 indicating a better model fit[61].

Illumina sequencing and analyses of 16S rRNA genes. 16S rRNA genes were amplified using the universal bacterial and archaeal primers, 515 F (5′-GTGCCAGCMGCCGCGGTAA-3′)[58] and 909 R (5′-CCCCGYCAATTCMTTTRAGT-3′)[62] with a unique 12-nucleotide barcode. The PCR mixture (25 µL) contained 12.5 µL of 2 × Taq MasterMix, 1 µL of each forward and reverse primer (10 µM), 9.5 µL of double-distilled H_2O, and 1 µL of DNA template. The PCR program included an initial denaturation at 94 °C for 3 min, followed by 30 cycles of denaturation at 94 °C for 40 s, annealing at 56 °C for 60 s, and extension at 72 °C for 60 s. A final extension was performed at 72 °C for 10 min. Two PCR reactions were run for each sample, and the amplified products were combined. The PCR products were purified with a DNA Gel Extraction Kit (Qingke). The purified PCR product was diluted, denatured, re-diluted, and mixed with PhiX (at 30% of the final DNA amount) as described in the Illumina library preparation protocols. The prepared library was then sequenced using paired-end 2 × 250 bp reads on an Illumina MiSeq system at the Chengdu Institute of Biology, Chinese Academy of Sciences.

The raw 16S rRNA gene sequence data were processed by QIIME2 Version 2019.4 (http:// docs.qiime2.org/2019.4/). All sequence reads were trimmed and assigned to each sample based on their unique barcodes. The high-quality sequences (> 150 bps in length with an average base quality score > 30) were selected for downstream analyses. The sequences were clustered as operational taxonomic units (OTUs) at a 97% identity threshold. The taxonomic classification of OTUs was annotated using the Classify-Sklearn Naève Bayes Taxonomy classifier in the feature-classifier plugin against the SILVA Release 138 Database.

PICRUSt2 was used to predict microbial functions based on 16S rRNA gene sequences, following the instructions of the developer[58]. The Nearest-Sequenced Taxon Index (NSTI) was calculated for each 16S marker sequence to evaluate its phylogenetic distance from reference genomes or sequences. Sequences with NSTI values greater than 2, typically representing uncharacterized phyla or off-target sequences, were excluded from the dataset. In this study, the weighted NSTI values of all sequences from the eight soils ranged from 0.31 to

0.39. Functional predictions were conducted by inferring Kyoto Encyclopedia of Genes and Genomes (KEGG) orthologs (KO) using PICRUSt2[63].

Metagenomic sequencing. Genomic DNA of cells from the SRB enrichments was extracted by a DNeasy PowerSoil Kit (QIAGEN), and the DNA samples were purified and prepared for metagenomic sequencing according to the procedures described above and the manufacturer's instructions. Paired-end sequencing (150 bp) was carried out using the Illumina NovaSeq platform (San Diego, CA). Raw sequences were initially trimmed by Trimmomatic[64] (v0.39) with a sliding window size of 4 bp. Screened reads were further trimmed from both ends using a minimum quality threshold of 3 bp, retained only if the average quality score exceeded Q15, and discarded if shorter than 80 bp. To remove human genome contamination, reads were aligned to the human reference genome (GRCh38) using Bowtie2[65] (v2.5.0) with the following parameters: -D 5 -R 1 -N 0 -L 22 -i S, 0, 2.50. Antibiotic resistance genes (ARGs) were identified and quantified using ARGs-OAP[66] (v3.0). Overall gene and pathway abundances were evaluated with HUMAnN[67,68] (v3.6). High-quality reads were assembled into contigs using metaSPAdes[69] (v3.13.0), and contig quality was assessed with QUAST[70] (v5.0.2). Gene prediction was performed using Prodigal[71] (v2.6.3), taxonomic classification with Kaiju[72] (v1.7.2), and protein annotation with eggNOG-mapper[73].

Initial binning was conducted using MetaWRAP[74] v1.3.2, following the procedure described by Zhao et al.[75]. Bins were retained if they exhibited genome completeness greater than 70% and contamination below 5%. Annotation and taxonomic classification of the bins were performed using Bakta[76] (v1.5.1) and GTDB-Tk[77] (v2.1.0), respectively.

In this study, SRBs were identified and classified based on the presence of sulfite dissimilatory reductase genes *dsrA* and *dsrB*[78]. Functional genes, including *dsrA*, *dsrB*, and those involved in the reductive acetyl-CoA pathway, were identified and quantified through homology-based alignment against the UniProt and KEGG databases using Diamond[79] (v2.0.14) with an E-value threshold of <0.00001.

Identification and quantification of *hgcAB* genes

The identification and quantification of *hgcAB* genes were performed based on the metagenomic data following the approach described by Xia et al.[80] Each metagenome-assembled genome (MAG) was first subjected to open reading frame (ORF) prediction using Prodigal[67]. Specifically, *hgcAB* genes were identified from predicted ORFs using a hidden Markov model (HMM)-based pipeline[81]. Candidate *hgcAB* sequences were retained if they met stringent criteria: an e-value ≤ 1 E − 50 or a score ≥ 300, with additional hits considered if the score was ≥ 70. High-confidence sequences were concatenated and aligned using MUSCLE[82] and visualized with AliView[83]. Sequences lacking the conserved cap-helix domain [G(I/V)NVWCAAGK] or the two strictly conserved ferredoxin-binding motifs (CXXCXXCXXXC) were excluded[13]. MAGs were classified as *hgcAB*-harboring if both *hgcA* and *hgcB* genes were detected within the same bin. The abundance of *hgcAB*-positive MAGs was estimated using the "quant_bins" module of MetaWRAP, based on the normalized coverage (TPM) of each MAG across treatments[81]. Additionally, a Bowtie2 index was constructed from the identified *hgcAB* sequences to further assess gene-level abundance[65]. Paired-end reads were mapped to this index using Bowtie2 with default parameters. The total coverage of each *hgcAB* sequence was calculated using the pileup.sh script from the BBMap toolkit (http://sourceforge.net/projects/bbmap/). The normalized abundance of each representative *hgcA* sequence was calculated as total coverage per gigabase (Gb) of metagenomic data, allowing direct comparison of *hgcAB* abundance across metagenomes of varying sizes.

Statistical analyses

Statistical analyses were conducted using R version 4.1.2 (www.r-project.org). Bar and box plots were generated using the 'ggplot2' package[84] (v3.4.1), and radar plots with the 'ggradar' package[85] (v0.2). Non-metric multidimensional scaling (NMDS) was based on Bray–Curtis dissimilarities of the microbial community using the vegan package[86] (v2.6-10). The 16S rRNA gene-based community was ordinated using operational taxonomic unit (OTU) abundances, while the *hgcA*-containing microbial community was ordinated using genus-level abundances. The significance of community separation among treatments was assessed using permutational multivariate analysis of variance (PERMANOVA). Venn diagrams were created using the 'VennDiagram' package[87] (v1.7.3). Partial Least Squares Path Modeling (PLS-PM) was carried out using the 'plspm' package[88] (v0.5.0), with model evaluation based on the methodology described by Latan[89]. The PLS-PM analysis focused on environmental factors that are strongly associated with MeHg synthesis. Significant differences were analyzed using the 'ggsignif' package[90] (v0.6.4) and the 'rstatix' package[91] (v0.7.2). Data normality was confirmed using the Shapiro-Wilk normality test. All plots were finalized and refined using Adobe Illustrator 2024.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data underlying the results presented in the manuscript and the Supplementary Information have been plotted in the figures or provided in the tables and are available as source data (https://doi.org/10.6084/m9.figshare.33416887). The DNA sequencing data generated in this study have been deposited in the NCBI GenBank database, including the 16S rRNA gene sequencing data (PRJNA1522449) and metagenomic sequencing data (PRJNA1524093). The *hgcA* sequencing data, including the assembled *Deltaproteobacteria-hgcA* and Archaea-*hgcA* sequences, have been deposited at https://doi.org/10.6084/m9.figshare.33439240.

Code availability

In this study, all data analyses were performed using custom R scripts and publicly available R packages. All custom code generated for the analyses is available upon request.

References

1. Drlica, K., Malik, M., Kerns, R. J. & Zhaol, X. L. Quinolone-mediated bacterial death. *Antimicrob. Agents Ch.* **52**, 385–392 (2008).
2. Pham, T. D. M., Ziora, Z. M. & Blaskovich, M. A. T. Quinolone antibiotics. *MedChemComm* **10**, 1719–1739 (2019).
3. Kampranis, S. C. & Maxwell, A. The DNA gyrase-quinolone complex - ATP hydrolysis and the mechanism of DNA cleavage. *J. Biol. Chem.* **273**, 22615–22626 (1998).
4. Zhang, Q. Q., Ying, G. G., Pan, C. G., Liu, Y. S. & Zhao, J. L. Comprehensive evaluation of antibiotics emission and fate in the river basins of China: source analysis, multimedia modeling, and linkage to bacterial resistance. *Environ. Sci. Technol.* **49**, 6772–6782 (2015).
5. Sukul, P., Lamshöft, M., Kusari, S., Zühlke, S. & Spiteller, M. Metabolism and excretion kinetics of ^{14}C-labeled and non-labeled difloxacin in pigs after oral administration, and antimicrobial activity of manure containing difloxacin and its metabolites. *Environ. Res.* **109**, 225–231 (2009).
6. Pan, M. & Chu, L. M. Occurrence of antibiotics and antibiotic resistance genes in soils from wastewater irrigation areas in the Pearl River Delta region, southern China. *Sci. Total Environ.* **624**, 145–152 (2018).

7. Pan, X., Qiang, Z. M., Ben, W. W. & Chen, M. X. Residual veterinary antibiotics in swine manure from concentrated animal feeding operations in Shandong Province, China. *Chemosphere* **84**, 695–700 (2011).
8. Guo, T. et al. Increased occurrence of heavy metals, antibiotics and resistance genes in surface soil after long-term application of manure. *Sci. Total Environ.* **635**, 995–1003 (2018).
9. Sun, J. T. et al. Antibiotic resistance genes (ARGs) in agricultural soils from the Yangtze River Delta, China. *Sci. Total Environ.* **740**, 140001 (2020).
10. Parente, C. E. T. et al. Fluoroquinolones in agricultural soils: multi-temporal variation and risks in Rio de Janeiro upland region. *Chemosphere* **219**, 409–417 (2019).
11. Wei, R. C. et al. Occurrence of 13 veterinary drugs in animal manure-amended soils in Eastern China. *Chemosphere* **144**, 2377–2383 (2016).
12. Grenni, P., Ancona, V. & Caracciolo, A. B. Ecological effects of antibiotics on natural ecosystems: a review. *Microchem. J.* **136**, 25–39 (2018).
13. Parks, J. M. et al. The genetic basis for bacterial mercury methylation. *Science* **339**, 1332–1335 (2013).
14. Compeau, G. C. & Bartha, R. Sulfate–reducing bacteria–principal methylators of mercury in anoxic estuarine sediment. *Appl. Environ. Microb.* **50**, 498–502 (1985).
15. Gilmour, C. C. et al. Mercury methylation by novel microorganisms from new environments. *Environ. Sci. Technol.* **47**, 11810–11820 (2013).
16. Choi, S. C., Chase, T. & Bartha, R. Metabolic pathways leading to mercury methylation in *Desulfovibrio-desulfuricans* Ls. *Appl. Environ. Microb.* **60**, 4072–4077 (1994).
17. Zheng, K. Y. et al. S- adenosyl- L- methionine is the unexpected methyl donor for the methylation of mercury by the membrane-associated *HgcAB* complex. *Proc. Natl. Acad. Sci. USA* **121**, e2408086121 (2024).
18. Yu, R. Q. & Barkay, T. Microbial mercury transformations: molecules, functions and organisms. *Adv. Appl. Microbiol.* **118**, 31–90 (2022).
19. Zhang, H., Song, S., Jia, Y., Wu, D. & Lu, H. Stress-responses of activated sludge and anaerobic sulfate-reducing bacteria sludge under long-term ciprofloxacin exposure. *Water Res.* **164**, 114964 (2019).
20. King, J. K., Kostka, J. E., Frischer, M. E. & Saunders, F. M. Sulfate-reducing bacteria methylate mercury at variable rates in pure culture and in marine sediments. *Appl. Environ. Microb.* **66**, 2430–2437 (2000).
21. Yu, R. Q., Reinfelder, J. R., Hines, M. E. & Barkay, T. Syntrophic pathways for microbial mercury methylation. *ISME J.* **12**, 1826–1835 (2018).
22. Córdova-Kreylos, A. L. & Scow, K. M. Effects of ciprofloxacin on salt marsh sediment microbial communities. *ISME J.* **1**, 585–595 (2007).
23. Fang, F. et al. Quinolone antibiotics stimulate bacterial mercury methylation by *Geobacter metallireducens* GS-15. *Bioresour. Technol.* **413**, 131465 (2024).
24. Qin, C. Y. et al. Isotopic fractionation and source appointment of methylmercury and inorganic mercury in a paddy ecosystem. *Environ. Sci. Technol.* **54**, 14334–14342 (2020).
25. Karagas, M. R. et al. Evidence on the human health effects of low-level methylmercury exposure. *Environ. Health Persp.* **120**, 799–806 (2012).
26. Dai, S. S. et al. Global distribution and environmental drivers of methylmercury production in sediments. *J. Hazard. Mater.* **407**, 124700 (2021).
27. Zhong, H. et al. Soil *Geobacteraceae* are the key predictors of neurotoxic methylmercury bioaccumulation in rice. *Nat. Food* **5**, 301–311 (2024).
28. Kerin, E. J. et al. Mercury methylation by dissimilatory iron-reducing bacteria. *Appl. Environ. Microb.* **72**, 7919–7921 (2006).
29. Windham-Myers, L. et al. Mercury cycling in agricultural and managed wetlands: a synthesis of methylmercury production, hydrologic export, and bioaccumulation from an integrated field study. *Sci. Total Environ.* **484**, 221–231 (2014).
30. Lei, P. et al. Understanding the effects of sulfur input on mercury methylation in rice paddy soils. *Sci. Total Environ.* **778**, 146325 (2021).
31. Wang, J. T., Dai, J., Chen, G. H. & Jiang, F. Role of sulfur biogeochemical cycle in mercury methylation in estuarine sediments: A review. *J. Hazard. Mater.* **423**, 126964 (2022).
32. Marvin-Dipasquale, M. C. & Oremland, R. S. Bacterial methylmercury degradation in Florida Everglades peat sediment. *Environ. Sci. Technol.* **32**, 2556–2563 (1998).
33. Hintelmann, H., Keppel-Jones, K. & Evans, R. D. Constants of mercury methylation and demethylation rates in sediments and comparison of tracer and ambient mercury availability. *Environ. Toxicol. Chem.* **19**, 2204–2211 (2000).
34. Hsu-Kim, H., Kucharzyk, K. H., Zhang, T. & Deshusses, M. A. Mechanisms regulating mercury bioavailability for methylating microorganisms in the aquatic environment: a critical review. *Environ. Sci. Technol.* **47**, 2441–2456 (2013).
35. Zhong, H. et al. Mercury supply limits methylmercury production in paddy soils. *Sci. Total Environ.* **927**, 172335 (2024).
36. Ragsdale, S. W. & Pierce, E. Acetogenesis and the Wood–Ljungdahl pathway of CO_2 fixation. *BBA-Proteins Proteom.* **1784**, 1873–1898 (2008).
37. Sah, S., Aluri, S., Rex, K. & Varshney, U. One-carbon metabolic pathway rewiring in reveals an evolutionary advantage of 10-Formyltetrahydrofolate Synthetase (*Fhs*) in survival under hypoxia. *J. Bacteriol.* **197**, 717–726 (2015).
38. Cycon, M., Mrozik, A. & Piotrowska-Seget, Z. Antibiotics in the soil environment-degradation and their impact on microbial activity and diversity. *Front. Microbiol.* **10**, 338 (2019).
39. Oliphant, C. M. & Green, G. M. Quinolones: a comprehensive review. *Am. Fam. Physician* **65**, 455–464 (2002).
40. Stein, G. E. & Goldstein, E. J. C. Fluoroquinolones and anaerobes. *Clin. Infect. Dis.* **42**, 1598–1607 (2006).
41. Jia, Y. Y. et al. Ciprofloxacin degradation in anaerobic sulfate-reducing bacteria (SRB) sludge system: Mechanism and pathways. *Water Res.* **136**, 64–74 (2018).
42. Miritana, V. M. et al. Effects of ciprofloxacin alone or in mixture with sulfamethoxazole on the efficiency of anaerobic digestion and its microbial community. *Antibiotics* **11**, 1111 (2022).
43. Hausmann, B. et al. Consortia of low-abundance bacteria drive sulfate reduction-dependent degradation of fermentation products in peat soil microcosms. *ISME J.* **10**, 2365–2375 (2016).
44. Yu, T. T. et al. Anaerobic degradation of organic carbon supports uncultured microbial populations in estuarine sediments. *Microbiome* **11**, 81 (2023).
45. Gilmour, C. C. et al. Methylmercury concentrations and production rates across a trophic gradient in the northern Everglades. *Biogeochemistry* **40**, 327–345 (1998).
46. Oremland, R. S. & Polcin, S. Methanogenesis and sulfate reduction–competitive and noncompetitive substrates in estuarine sediments. *Appl. Environ. Microb.* **44**, 1270–1276 (1982).
47. Jin, H. et al. Distribution pattern analysis of multiple antibiotics in the soil-rice system using a QuEChERS extraction method. *Sci. Total Environ.* **944**, 173929 (2024).
48. Morales-Muñoz, S., Luque-García, J. L. & de Castro, M. D. L. Continuous microwave-assisted extraction coupled with derivatization and fluorimetric monitoring for the determination of fluoroquinolone antibacterial agents from soil samples. *J. Chromatogr. A* **1059**, 25–31 (2004).

49. Lei, P. et al. Properly interpret metabolic inhibition results to identify primary mercury methylating microbes. *Crit. Rev. Env. Sci. Tec.* **53**, 1757–1773 (2023).
50. Shan, X. Q. et al. Relationship between sulfur speciation in soils and plant availability. *Sci. Total Environ.* **199**, 237–246 (1997).
51. Chen, C. et al. Sulfate-reducing bacteria and methanogens are involved in arsenic methylation and demethylation in paddy soils. *ISME J.* **13**, 2523–2535 (2019).
52. Atlas R. M. *Handbook of media for environmental microbiology*, 2nd edn. (Taylor & Francis, 2005).
53. Carrasco, L., Díez, S. & Bayona, J. M. Methylmercury determination in biota by solid-phase microextraction matrix effect evaluation. *J. Chromatogr. A* **1174**, 2–6 (2007).
54. Tang, W. L., Chu, J. R. & Zhong, H. Alkaline extraction: an optimal approach for extracting methylmercury from paddy soils. *Sci. Total Environ.* **885**, 163776 (2023).
55. Guo, Y. Y. et al. Trojan Horse" type internalization increases the bioavailability of mercury sulfide nanoparticles and methylation after intracellular dissolution. *ACS Nano* **17**, 1925–1934 (2023).
56. Turiel, E., Martín-Esteban, A. & Tadeo, J. L. Multiresidue analysis of quinolones and fluoroquinolones in soil by ultrasonic-assisted extraction in small columns and HPLC-UV. *Anal. Chim. Acta* **562**, 30–35 (2006).
57. Christensen, G. A. et al. Development and validation of broad-range qualitative and clade-specific quantitative molecular probes for assessing mercury methylation in the environment. *Appl. Environ. Microb.* **82**, 6068–6078 (2016).
58. Tuan, N. N., Chang, Y. C., Yu, C. P. & Huang, S. L. Multiple approaches to characterize the microbial community in a thermophilic anaerobic digester running on swine manure: a case study. *Microbiol. Res.* **169**, 717–724 (2014).
59. Bravo, A. G. et al. Methanogens and iron-reducing bacteria: the overlooked members of mercury-methylating microbial communities in Boreal Lakes. *Appl. Environ. Microb.* **84**, e01774–18 (2018).
60. Liu, Y. R. et al. Unraveling microbial communities associated with methylmercury production in paddy soils. *Environ. Sci. Technol.* **52**, 13110–13118 (2018).
61. Henseler, J. & Sarstedt, M. Goodness-of-fit indices for partial least squares path modeling. *Comput. Stat.* **28**, 565–580 (2013).
62. Beckers, B. et al. Performance of 16s rDNA primer pairs in the study of rhizosphere and endosphere bacterial microbiomes in metabarcoding studies. *Front. Microbiol.* **7**, 650 (2016).
63. Douglas, G. M. et al. PICRUSt2 for prediction of metagenome functions. *Nat. Biotechnol.* **38**, 685–688 (2020).
64. Bolger, A. M., Lohse, M. & Usadel, B. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* **30**, 2114–2120 (2014).
65. Langmead, B. & Salzberg, S. L. Fast gapped-read alignment with Bowtie 2. *Nat. Methods* **9**, 357–U354 (2012).
66. Yin, X. L. et al. ARGs-OAP v3.0: Antibiotic-resistance gene database curation and analysis pipeline optimization. *Eng.-Prc.* **27**, 234–241 (2023).
67. Abubucker, S. et al. Metabolic reconstruction for metagenomic data and its application to the human microbiome. *PLoS Comput. Biol.* **8**, e1002358 (2012).
68. Beghini, F. et al. Integrating taxonomic, functional, and strain-level profiling of diverse microbial communities with bioBakery 3. *Elife* **10**, e65088 (2021).
69. Nurk, S., Meleshko, D., Korobeynikov, A. & Pevzner, P. A. metaSPAdes: a new versatile metagenomic assembler. *Genome Res.* **27**, 824–834 (2017).
70. Manchanda, N. et al. GenomeQC: a quality assessment tool for genome assemblies and gene structure annotations. *BMC Genomics* **21**, 193 (2020).
71. Hyatt, D. et al. Prodigal: prokaryotic gene recognition and translation initiation site identification. *BMC Bioinforma.* **11**, 119 (2010).
72. Menzel, P., Ng, K. L. & Krogh, A. Fast and sensitive taxonomic classification for metagenomics with Kaiju. *Nat. Commun.* **7**, 11257 (2016).
73. Cantalapiedra, C. P., Hernández-Plaza, A., Letunic, I., Bork, P. & Huerta-Cepas, J. eggNOG-mapper v2: functional annotation, orthology assignments, and domain prediction at the metagenomic scale. *Mol. Biol. Evol.* **38**, 5825–5829 (2021).
74. Uritskiy, G. V., DiRuggiero, J. & Taylor, J. MetaWRAP-a flexible pipeline for genome-resolved metagenomic data analysis. *Microbiome* **6**, 158 (2018).
75. Zhao, R. X. et al. Deciphering the mobility and bacterial hosts of antibiotic resistance genes under antibiotic selection pressure by metagenomic assembly and binning approaches. *Water Res.* **186**, 116318 (2020).
76. Schwengers, O. et al. Bakta: rapid and standardized annotation of bacterial genomes via alignment- free sequence identification. *Microb. Genomics.* **7**, 685 (2021).
77. Chaumeil, P. A., Mussig, A. J., Hugenholtz, P. & Parks, D. H. GTDB-Tk v2: memory friendly classification with the genome taxonomy database. *Bioinformatics* **38**, 5315–5316 (2022).
78. Magnuson, E., Altshuler, I., Freyria, N. J., Leveille, R. J. & Whyte, L. G. Sulfur-cycling chemolithoautotrophic microbial community dominates a cold, anoxic, hypersaline Arctic spring. *Microbiome* **11**, 203 (2023).
79. Buchfink, B., Xie, C. & Huson, D. H. Fast and sensitive protein alignment using DIAMOND. *Nat. Methods* **12**, 59–60 (2015).
80. Xia, J. T., Yuan, Z. G. & Jiang, F. Global metagenomic survey identifies sewage-derived microorganisms as key contributors to riverine methylmercury production. *Nat. Commun.* **15**, 9262 (2024).
81. McDaniel, E. A. et al. Expanded phylogenetic diversity and metabolic flexibility of mercury-methylating microorganisms. *Msystems* **5**, e00299–20 (2020).
82. Edgar, R. C. MUSCLE: multiple sequence alignment with high accuracy and high throughput. *Nucleic Acids Res.* **32**, 1792–1797 (2004).
83. Larsson, A. AliView: a fast and lightweight alignment viewer and editor for large datasets. *Bioinformatics* **30**, 3276–3278 (2014).
84. Wickham H., Sievert C. *ggplot2: elegant graphics for data analysis*. 2nd edn. (Springer, 2016).
85. Moon K.-W. *Learn ggplot2 using shiny App*. 1st edn. (Springer, 2016).
86. Dixon, P. VEGAN, a package of R functions for community ecology. *J. Veg. Sci.* **14**, 927–930 (2003).
87. Chen, H. & Boutros, P. C. VennDiagram: a package for the generation of highly-customizable Venn and Euler diagrams in R. *BMC Bioinforma.* **12**, 35 (2011).
88. Davino, C. & Vinzi, V. E. Quantile composite-based path modeling. *Adv. Data Anal. Cl.* **10**, 491–520 (2016).
89. Latan H., Latan H., Noonan R. *Partial Least Squares Path Modelin: basic concepts, methodological issues and applications*. 1st edn. (Springer, 2017).
90. Patil; CA-E. I. *ggsignif: R package for displaying significance brachgckets for 'ggplot2'*. (GGPLOT, 2022).
91. Kassambara A. *rstatix: pipe-friendly framework for basic statistical tests. Comprehensive R Archive Network* (CRAN, 2023).

Acknowledgements

In this study, we thank Zizhu Li, Yu Liang, Yasa Wang and Honghong Li for help with collection of soil samples. We also appreciate Minjie Yao and Chi Liu from Fujian Agriculture and Forestry University for help with the gene sequencing, and the technical assistance from Liuyan Yang, Wanxuan Dai, Chenghao Ge, Chao Wu, Xiaopeng Zhao, and Xiru Chen at Nanjing University.

Author contributions

H.L., W.T., H.Z., L.X., and R.-Q.Y. conceived the study. H.Z. and W.T. secured funding and supervised the research. H.L., R.-Q.Y., X.L., and J.X. generated all main analyses and figures. H.L., X.Z., M.W., J.L., Y.-N.Y., Y.L., H.-R.Z., and Y.H. performed the lab work. H.L. drafted the manuscript, and all the authors jointly revised the manuscript and figures, and R.-Q.Y., X.L., C.R., W.T., X.W., Y.L., and H.Z. approved the final version.

Funding

This work was supported by the National Natural Science Foundation of China (42525710 and 42577256) and Natural Science Foundation of Jiangsu Province (BK20230082).

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-026-79083-7.

Correspondence and requests for materials should be addressed to Wenli Tang, Ri-Qing Yu, Lin Xiao or Huan Zhong.

Peer review information *Nature Communications* thanks Simon Guerrero-Cruz, and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

© The Author(s) 2026

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.

[1]State Key Laboratory of Water Pollution Control and Green Resource Recycling, School of Environment, Nanjing University, Nanjing, China. [2]Engineering Research Center of Soil Remediation of Fujian Province University, College of Resources and Environment, Fujian Agriculture and Forestry University, Fuzhou, China. [3]Shenzhen Key Laboratory of Marine Microbiome Engineering, Institute for Advanced Study, Shenzhen University, Shenzhen, China. [4]Department of Biology, Center for Environment, Biodiversity and Conservation, University of Texas at Tyler, Tyler, TX, USA. [5]Institute of Environmental Microbiology, College of Resources and Environment, Fujian Agriculture and Forestry University, Fuzhou, China. [6]School of Environmental Science and Engineering, Sun Yat-sen University, Guangzhou, China. [7]Institute of Environmental Research at Greater Bay Area, Key Laboratory for Water Quality and Conservation of the Pearl River Delta, Ministry of Education, Guangzhou University, Guangzhou, China. ✉e-mail: tangwenli@nju.edu.cn; ryu@uttyler.edu; xiaolin@nju.edu.cn; zhonghuan@nju.edu.cn